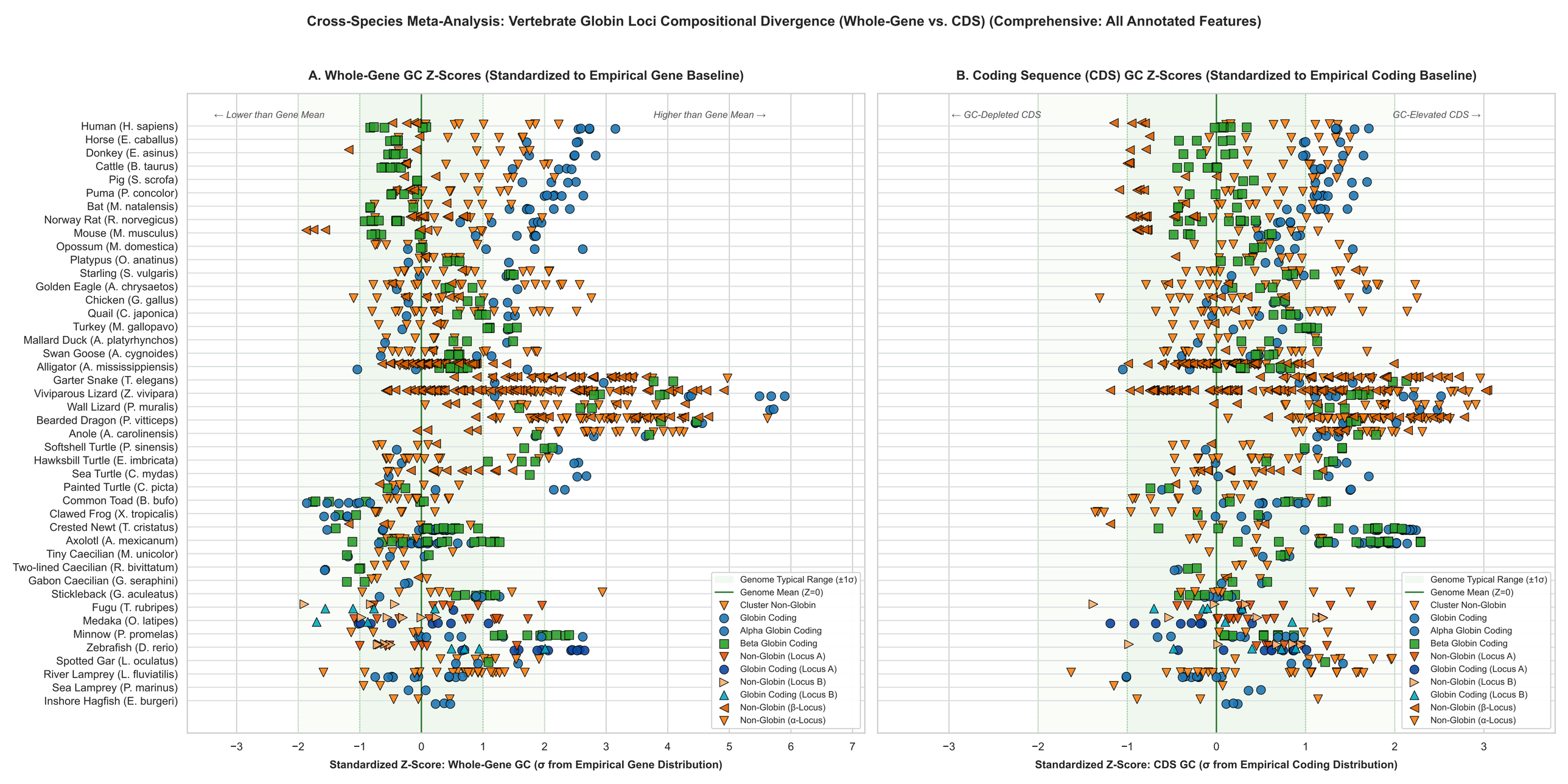

Cross-Species Meta-Analysis: Vertebrate Globin Loci Compositional Divergence (Whole-Gene vs. CDS) (Comprehensive: All Annotated Features)
A. Whole-Gene GC Z-Scores (Standardized to Empirical Gene Baseline)
B. Coding Sequence (CDS) GC Z-Scores (Standardized to Empirical Coding Baseline)
← Lower than Gene Mean
Higher than Gene Mean →
← GC-Depleted CDS
GC-Elevated CDS →
Human (H. sapiens)
Horse (E. caballus)
Donkey (E. asinus)
Cattle (B. taurus)
Pig (S. scrofa)
Puma (P. concolor)
Bat (M. natalensis)
Norway Rat (R. norvegicus)
Mouse (M. musculus)
Opossum (M. domestica)
Platypus (O. anatinus)
Starling (S. vulgaris)
Golden Eagle (A. chrysaetos)
Chicken (G. gallus)
Quail (C. japonica)
Turkey (M. gallopavo)
Mallard Duck (A. platyrhynchos)
Swan Goose (A. cygnoides)
Alligator (A. mississippiensis)
Garter Snake (T. elegans)
Viviparous Lizard (Z. vivipara)
Wall Lizard (P. muralis)
Bearded Dragon (P. vitticeps)
Anole (A. carolinensis)
Softshell Turtle (P. sinensis)
Hawksbill Turtle (E. imbricata)
Sea Turtle (C. mydas)
Painted Turtle (C. picta)
Common Toad (B. bufo)
Clawed Frog (X. tropicalis)
Crested Newt (T. cristatus)
Axolotl (A. mexicanum)
Tiny Caecilian (M. unicolor)
Two-lined Caecilian (R. bivittatum)
Gabon Caecilian (G. seraphini)
Stickleback (G. aculeatus)
Fugu (T. rubripes)
Medaka (O. latipes)
Minnow (P. promelas)
Zebrafish (D. rerio)
Spotted Gar (L. oculatus)
River Lamprey (L. fluviatilis)
Sea Lamprey (P. marinus)
Inshore Hagfish (E. burgeri)
Genome Typical Range (±1σ)
Genome Mean (Z=0)
Cluster Non-Globin
Globin Coding
Alpha Globin Coding
Beta Globin Coding
Non-Globin (Locus A)
Globin Coding (Locus A)
Non-Globin (Locus B)
Globin Coding (Locus B)
Non-Globin (β-Locus)
Non-Globin (α-Locus)
−3
−2
−1
0
1
2
3
4
5
6
7
Standardized Z-Score: Whole-Gene GC (σ from Empirical Gene Distribution)
Standardized Z-Score: CDS GC (σ from Empirical Coding Distribution)